MARIANO MARCOS STATE UNIVERSITY

QS STARS™ RATING SYSTEM

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RANK 2 IN THE PHILIPPINES

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RANK 356 IN THE WORLD

The Game Score Case

A small gaming company is testing a new tabletop game for students. In one round of the game, a player can get any one of the following four scores**: 2, 4, 6, and 8 points. Each score is equally likely to occur**, so these values represent the **population of possible scores.** To study how averages behave, the company considers all possible samples of size 2, taken without replacement, and computes the mean score for each sample. These sample means are then used to form the sampling distribution of the sample mean.

Answer the following:

a. Find the mean, variance, and standard deviation of the population.
b. List all possible samples of size 2 and find the mean of each sample.
c. Construct the sampling distribution of the sample means.
d. Find the mean, variance, and standard deviation of the sampling distribution of the sample means.

Solution

a. Mean, variance, and standard deviation of the population

Population values: 2, 4, 6, 8

Mean

$$\mu = \frac{2 + 4 + 6 + 8}{2} = \frac{20}{4} = 5$$

Variance

$$\sigma^2 = \frac{\sum(x - \mu)^2}{N}$$

$$(2 - 5)^2 = 9$$

$$(4 - 5)^2 = 1$$

$$(6 - 5)^2 = 1$$

$$(8 - 5)^2 = 9$$

$$\sum(x - \mu)^2 = 9 + 1 + 1 + 9 = 20$$

$$\sigma^2 = \frac{20}{4} = 5$$

Standard deviation

= 2.24

Solution

b. List all possible samples of size 2 and find the mean of each sample.

Sample	Mean
(2, 4)	3
(2, 6)	4
(2, 8)	5
(4, 6)	5
(4, 8)	6
(6, 8)	7

Solution

c. Construct the sampling distribution of the sample means.

Sample Mean	Frequency	Probability
3	1	1/6
4	1	1/6
5	2	2/6
6	1	1/6
7	1	1/6

Solution

d. Find the mean, variance, and standard deviation of the sampling distribution of the sample means.

The sample means are : 3, 4, 5, 5, 6, 7

Mean of the sampling distribution

$$\mu_{\bar{x}} = \frac{3+4+5+5+6+7}{6} = \frac{30}{6} = 5$$

Variance

$$\sigma_{\bar{x}}^2 = \frac{\sum(\bar{x}-\mu_{\bar{x}})^2}{n}$$

$$(3-5)^2 = 4$$
$$(4-5)^2 = 1$$
$$(5-5)^2 = 0$$
$$(5-5)^2 = 0$$
$$(6-5)^2 = 1$$
$$(7-5)^2 = 4$$

$$\sum(\bar{x}-\mu_{\bar{x}})^2 = 4+1+0+0+1+4 = 10$$

$$\sigma_{\bar{x}}^2 = \frac{10}{6} \approx 1.67$$

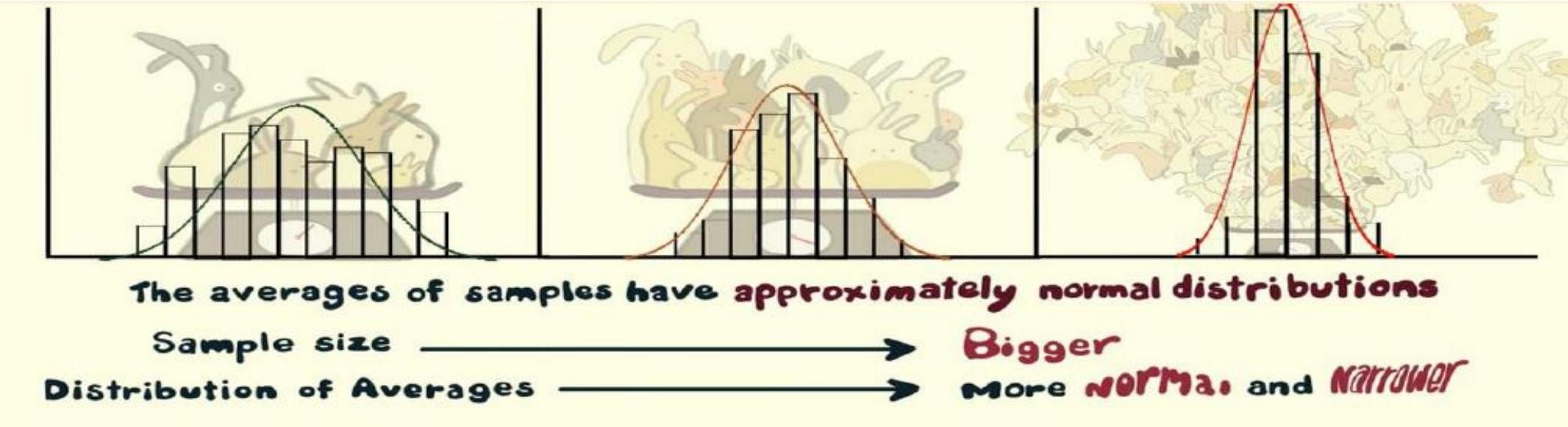
Standard deviation

CENTRAL LIMIT THEOREM

CMPSC 136 Quantitative Methods for Computer Science

The Central Limit Theorem

If the sample size n is sufficiently large, then the population od all possible sample means is approximately normally distributed mean ($\mu_{\overline{X}} = \mu$) and standard deviation ($\sigma_{\overline{X}} = \sigma/\sqrt{n}$), no matter what probability distribution describes the sampled population. Furthermore, the larger the sample size *n* is, the more nearly normally distributed is the population of all possible sample means.



The Central Limit Theorem

For samples of size 30 or more, the sample mean is approximately normally distributed, with mean $\mu_{\overline{X}} = \mu$ and the standard deviation $\sigma_{\overline{X}} = \frac{\sigma}{\sqrt{n}}$.

The Central Limit Theorem

Standard Error

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

σ - population standard deviation
n - sample size

Z-score Formula

$$z = \frac{\bar{x} - \mu}{\sigma_{\bar{x}}}$$

μ (mu) - population mean
$\bar{x}$ (x-bar) - sample mean
$\sigma_{\bar{x}}$ - standard error

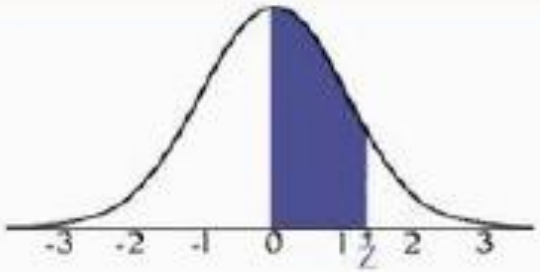
Exercise

Let $\bar{x}$ be the mean of a random sample of size 60 drawn from a population with a mean of 120 and a standard deviation of 45.

a)Find the probability that $\bar{x}$ assumes a value between 112 and 125.

b)Find the probability that $\bar{x}$ assumes a value greater than 117.

c)Find the probability that $\bar{x}$ assumes a value less than 110.



STANDARD NORMAL TABLE (Z)

Entries in the table give the area under the curve between the mean and z standard deviations above the mean. For example, for $z = 1.25$ the area under the curve between the mean (0) and z is 0.3944.

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0190	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1628	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2517	0.2549
0.7	0.2580	0.2611	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2881	0.2910	0.2939	0.2969	0.2995	0.3023	0.3051	0.3078	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3289	0.3315	0.3340	0.3365	0.3389
1.0	0.3413	0.3438	0.3461	0.3485	0.3508	0.3513	0.3554	0.3577	0.3529	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3907	0.3925	0.3944	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4147	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4279	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4463	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4616	0.4625	0.4633
1.8	0.4641	0.4649	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4761	0.4767
2.0	0.4772	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.1	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.2	0.4861	0.4864	0.4868	0.4871	0.4875	0.4878	0.4881	0.4884	0.4887	0.4890
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4913	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4932	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4987	0.4987	0.4987	0.4988	0.4988	0.4989	0.4989	0.4989	0.4990	0.4990
3.1	0.4990	0.4991	0.4991	0.4991	0.4992	0.4992	0.4992	0.4992	0.4993	0.4993
3.2	0.4993	0.4993	0.4994	0.4994	0.4994	0.4994	0.4994	0.4995	0.4995	0.4995
3.3	0.4995	0.4995	0.4995	0.4996	0.4996	0.4996	0.4996	0.4996	0.4996	0.4997
3.4	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4997	0.4998

Assignment

Let $\bar{x}$ be the mean of a random sample of size 75 drawn from a population with a mean of 150 and a standard deviation of 50.

a)Find the probability that $\bar{x}$ assumes a value between 100 and 155.

b)Find the probability that $\bar{x}$ assumes a value greater than 120.

c)Find the probability that $\bar{x}$ assumes a value less than 105.

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